***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 9 Microbial Systems Biology

9.1 Multiple Choice Questions

1) ________ of prokaryotic genomes are now available in public databases.
A) Dozens
B) Hundreds
C) Thousands
D) Millions
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.1

2) Based on the table of average intron frequency given below, predict the relative order of genome size for these four eukaryotic organisms.

Organism	Average Intron Frequency
Cryptosporidium parvum	0
Plasmodium falciparum	1
Arabidopsis thaliana	5
Homo sapiens	8

A) *Homo sapiens* > *Arabidopsis thaliana* > *Plasmodium falciparum* > *Cryptosporidium parvum*
B) *Cryptosporidium parvum* > *Plasmodium falciparum* > *Arabidopsis thaliana* > *Homo sapiens*
C) *Homo sapiens* > *Cryptosporidium parvum* > *Plasmodium falciparum* > *Arabidopsis thaliana*
D) Intron frequency cannot be used to predict genome size in eukaryotes.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.6

3) The 140 kbp genome containing many short repeats, ribosomal RNA genes, one RNA polymerase gene (*rpo*) and one RubisCO gene (*rbcL*) is most likely from a(n)
A) plant.
B) autotrophic bacterium.
C) chloroplast.
D) autotrophic archaeon.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.4

4) The evolutionary origin of mitochondria is demonstrated by which of the following key genomic characteristics?
A) variable numbers of protein-encoding genes
B) large stretches of AT-rich DNA
C) linear and circular genome structure in different organisms
D) rRNA genes most closely related to *Bacteria*
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.4

5) An open reading frame (ORF) encodes for
A) a carbohydrate.
B) a polypeptide.
C) a lipid.
D) a carbohydrate or a polypeptide.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

6) The pan genome of a species is the genomic content that is
A) common to all strains of the same species.
B) present in one or more strains of the same species.
C) shared with all other prokaryotes.
D) hypothetical or uncharacterized genome content of a species.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.7

7) Microarrays can be used to
A) analyze global gene expression.
B) detect pathogens.
C) detect unwanted food additives or substitutes.
D) detect pathogens, analyze global gene expression, and detect unwanted food additives or substitutes.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.9

8) Transcriptome analysis is useful in relation to genome analysis because
A) it is NOT dependent on nucleic acid sequencing technology.
B) it results in amino acid sequence and is, thus, easier to analyze.
C) it analyzes RNA, thus it reveals which genes are expressed under different conditions.
D) it reveals interactions between molecules and, thus, provides more information than genome analysis.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.9

9) The most abundant genes in prokaryotic genomes are
A) those involved in metabolism.
B) those involved in translation.
C) those involved in transport.
D) those involved in DNA replication.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

10) When compared with *Bacteria*, species of *Archaea* seem to devote a higher percentage of their genomes to genes encoding proteins involved in
A) transcription.
B) energy and coenzyme production.
C) cell membrane functions.
D) carbohydrate metabolism.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

11) The entire complement of RNA produced under a given set of conditions is called a(n)
A) array.
B) genome.
C) proteome.
D) transcriptome.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.9

12) The very first DNA sequencing technology called the Sanger method relies on
A) nanopore technology that separates DNA molecules based on charge differences.
B) the incorporation of dideoxynucleotides that terminate chain extension during DNA synthesis.
C) nuclear magnetic resonance (NMR) analysis.
D) the release of protons whenever a new nucleotide is added to a growing strand of DNA.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

13) The advantage(s) of second-generation DNA sequencing compared to the Sanger method are the result of
A) miniaturization of reaction size.
B) increased computing power.
C) increased length of DNA sequences obtained.
D) miniaturization of reaction size and increased computer power.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

14) Within *Archaea* and *Bacteria*, one megabase pair (Mbp) of DNA encodes about ________ open reading frames.
A) 10
B) 100
C) 1,000
D) 10,000
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

15) You want to know whether the virulence genes present in *Bordetella pertussis* are evolutionarily related to genes in the less pathogenic species *B. bronchiseptica* or if the virulence genes were acquired via horizontal gene transfer. What characteristic(s) would you compare to answer this question?
A) percentage of GC content and codon usage
B) genome size and number of introns
C) number of genes in the pan genome
D) ribosomal binding site and intron sequence
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.5

16) The science that applies powerful computational tools to DNA and protein sequences for the purpose of analyzing, storing, and accessing the sequences for comparative purposes is known as
A) metagenomics.
B) proteomics.
C) bioinformatics.
D) genomics.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

17) Most mobile DNA consists of
A) transposable elements.
B) introns.
C) linear chromosomes.
D) plasmids.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.7

18) Mobile DNA elements are more common in the genomes of
A) *Archaea*.
B) hyperthermophiles.
C) rapidly evolving species.
D) pathogens.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.7

19) Chromosomal islands contain clusters of genes for
A) DNA repair and replication.
B) virulence, biodegradation of pollutants, and symbiotic relationships.
C) catabolic and anabolic reactions.
D) antibiotic resistance.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.7

20) A computer program recognizes an ORF by looking for ribosomal binding sites, start codons, and stop codons with an appropriate number of nucleotides between each element. What is a drawback of this approach?
A) Too many ORFS are identified, most of which are stretches on non-coding junk DNA.
B) Codon bias causes incorrect annotations.
C) Legitimate genes and non-coding RNA may be missed.
D) We lack the computing power to complete the analyses in a timely manner, thus many genomes are only partially annotated.
Answer: C
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.2

21) The total genetic complement of all cells within a microbial community is called a(n)
A) chromosomal island.
B) interactome.
C) metagenome.
D) metabolome.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.8

22) Linking an ORF with a specific function is an example of gene
A) annotation.
B) assembly.
C) codon bias.
D) expression.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

23) The first genome sequenced was that of a(n)
A) virus.
B) bacterium.
C) eukaryote.
D) archaeon.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.1

24) In general, prokaryotic genome size is linked to its metabolism and interaction with the environment. Which type of prokaryote typically has the smallest genome?
A) parasite
B) endosymbiont
C) autotroph
D) free-living heterotroph
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

25) Chloroplasts and mitochondria originated from ________ by a process known as ________.
A) bacteria / chromosome reduction
B) plants / gene deletion
C) bacteria / endosymbiosis
D) plants / endosymbiosis
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.4

26) In addition to chromosomes, some mitochondria also contain other genetic material known as
A) plasmids.
B) transposons.
C) proteomes.
D) lysosomes.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.4

27) Gene function is annotated based on homology between the ORFs of a genome and proteins whose function has been proven experimentally. What ″-omic″ approach could help us determine the function and structure of proteins encoded by uncharacterized ORFs?
A) metabolomics
B) metagenomics
C) transcriptomics
D) proteomics
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.10

28) Which method is most commonly used in metabolomics?
A) ion torrent semiconductor sequencing
B) mass spectrometry
C) 2D polyacrylamide gel electrophoresis
D) Sanger method
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.11

29) Many of the genes in the chloroplast genome encode proteins involved in ________, whereas mitochondrial genomes primarily encode proteins involved in ________.
A) autotrophy / DNA replication
B) photosynthesis / glycolysis
C) autotrophy / glycolysis
D) photosynthesis / oxidative phosphorylation
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.4

30) Genome assembly relies on
A) accurate gene annotation.
B) overlap of a large numbers of short sequences.
C) codon bias.
D) systems biology.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

31) The core genome of a prokaryotic species is the genomic content that is
A) unique compared to all other prokaryotes.
B) expressed under all environmental conditions.
C) shared between all other prokaryotes.
D) shared between all strains of a species.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.7

32) Genes encoding compounds such as pyruvate, acetyl-CoA, fructose-6-phosphate, oxaloacetate, and other small organic compounds could be part of a(n)
A) metagenome.
B) metabolome.
C) interactome.
D) transcriptome.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.11

33) Genomes of species in both *Bacteria* and *Archaea* show a strong correlation between genome size and
A) noncoding RNA.
B) codon bias.
C) number of ORFs.
D) number of introns.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

34) Typically ________ of ORFs in a genome cannot clearly be assigned a function. These ORFs are usually assigned and predicted to encode hypothetical proteins.
A) < 1%
B) 5%
C) 30%
D) 90%
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

35) Fourth-generation sequencing methods no longer use ________ to detect nucleotide incorporation.
A) mass
B) pH
C) charge
D) light
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

36) RNA-Seq analysis is a method aimed at defining a(n)
A) metabolome.
B) transcriptome.
C) interactome.
D) metagenome.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.9

37) Genes from different sources that are related in sequence due to shared evolutionary ancestry are called ________ genes, and groups of such genes are known as ________.
A) homologous / gene families
B) paralogous / functional genes
C) orthologous / gene families
D) homologous / functional genes
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.5

38) Horizontal gene transfer
A) is rare and only occurs between closely related strains.
B) is common and may sometime occur between unrelated organisms.
C) does not provide an advantage to organisms.
D) only occurs in prokaryotes.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.6

39) Horizontal gene transfer has been thoroughly documented for genes involved in
A) DNA replication and repair.
B) virulence and metabolic functions.
C) only very few processes.
D) translation.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.6

40) A surprising finding of environmental metagenomic studies is that a majority of genes in the environment are
A) from microbes grown in the lab.
B) viral in origin.
C) eukaryotic in origin.
D) from extinct microorganisms.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.8

41) Which "omic" is most directly involved in identifying molecules as possible drug targets that can be used to develop new medications?
A) proteomics
B) genomics
C) transcriptomics
D) metabolomics
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.1

42) Why has research in metabolomics been slower than in genomics, transcriptomics, and proteomics?
A) Screening for metabolites is technically challenging due to their diversity.
B) Procedures have not yet been developed for studying the metabolome.
C) The molecules studied in metabolomics are too small for easy analysis.
D) Individual cells are too small to allow metabolomic analysis using existing techniques.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.11

9.2   True/False Questions

1) Knowledge of an organism's genome sequence yields important clues to how an organism functions and its evolutionary history.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.5

2) Genomic analysis led to the discovery that pathogenic organisms often lack genes for amino acid biosynthesis.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

3) Comparative genomics helps us to understand evolutionary relationships between organisms.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

4) Reconstructing evolutionary relationships helps in differentiating between primitive and derived characteristics.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.5

5) It is generally accepted that independent mutation rather than gene duplication is the mechanism for evolution of most new genes.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.5

6) Gene families are composed of homologous genes that have different evolutionary origins but the same function.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.5

7) Based on the evolutionary tree shown below, RubisCO Form II and RLP beta are orthologs.

```
RLP alpha
|                                            RLP beta
Ancestral gene______________________________|
|                                RLP ancestor
                      |          |          |
                      |          |
_____________________Duplicate gene
________________________________|___________RubisCO Form II
________________________________|___________|
________________________________RubisCO ancestor          RubisCO Form I
____________________________________________|_____________|
____________________________________________RubisCO Duplicate
__________________________________________________________|
__________________________________________________________RubisCO Form III
```

Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.5

8) Despite having smaller genomes, the protozoans *Paramecium* and *Trichomonas* have significantly more genes than humans.
Answer: TRUE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.4

9) Paralogs always have the same function.
Answer: FALSE
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.5

10) The key characteristic of third-generation sequencing is the ability to sequence single molecules of DNA.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

11) The largest cellular genomes belong to prokaryotes that are parasitic or pathogenic.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

12) Codon usage varies from one organism to the next.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

13) Genes for DNA replication and transcription make up only a small fraction of the typical prokaryotic genome.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

14) The relative percentage of genes devoted to protein synthesis in small-genome organisms is high compared with that of large-genome organisms.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

15) Few genes in all organisms have common evolutionary roots.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.5

16) Horizontally transferred genes typically encode essential metabolic functions such as DNA replication, transcription, and translation.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.6

17) Chromosomal rearrangements due to insertion sequences have apparently contributed to the evolution of several human pathogens.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.6

18) Some virulence genes are carried on plasmids or by lysogenic bacteriophages.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.7

19) Microorganisms that grow in extreme environments typically contain larger genomes when compared to microbes that grow in non-extreme environments.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

20) With modern molecular techniques, it is now possible to completely assemble a genome from a single cell.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.12

21) Most chloroplast genes are those responsible for photosynthesis and other demands of autotrophy.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.4

9.3 Essay Questions

1) After transcription, mRNA may undergo significant editing. Compare and contrast RNA editing in prokaryotes and eukaryotes and connect how these differences affect genome size and gene content.
Answer: RNA editing occurs after transcription of mRNA. Introns are pieces of RNA that are removed after transcription and before translation. Eukaryotic mRNA undergoes extensive RNA editing after transcription and in general eukaryotic genes contain many more introns than do prokaryotic genes. The number of introns per gene in eukaryotic genes ranges from 0-10, while most prokaryotic genes do not have any. The number of introns per gene is positively correlated with genome size. As the number of introns per gene increases, the genome size increases. The number of ORFs per 1000 nucleotides thus tends to decrease. In general, prokaryotic genomes encode for more genes (ORFs) per every 1000 base due to the lack of introns and less regulatory DNA.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.3

2) You are interested in the minimum set of genes necessary for survival of a eukaryotic microorganism such as *Saccharomyces cerevisiae*. Design an experiment to systematically test which genes are essential for survival and which are not under high nutrient, aerobic conditions.
Answer: The genome of *S. cerevisiae* can be either haploid or diploid. Genes can be systematically mutated or disabled (knocked out) in *S. cerevisiae* in the diploid state. Then each mutant can be tested for viability in the haploid state. Any mutant that dies in the haploid state had a mutation in a critical survival gene. Any mutant that survives in the haploid state had a mutation in a gene unnecessary for survival.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.3

3) Explain how codon bias and GC content can be used to detect horizontal gene transfer within a genome.
Answer: If an organism rarely uses a particular amino acid and one gene sequence has significantly more codons for the otherwise rare amino acid, this codon bias can be used to infer horizontal transfer of the gene. GC content may also be significantly different in a gene that has been transferred horizontally compared to the average GC content of the majority of the genome.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.6

4) Explain horizontal gene transfer and demonstrate how this phenomenon has complicated evolutionary studies using a diagram that illustrates phylogenetic relationships between organisms and genes.
Answer: Horizontal (or lateral) gene transfer involves a gene sequence being transferred from one cell to another non-progenic cell. This mechanism explains how one cell lineage acquires genes from others but complicates phylogenetics. It is therefore difficult to determine the evolutionary relatedness of organisms when they each contain a mosaic of genes laterally transferred from several different organisms.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.6

5) Why are the words ″about″ and ″approximately″ used in discussing the results of genomic analysis? Predict what advances in knowledge or methods could change our understanding of a genome sequence.
Answer: As genome sequence analysis becomes more refined, the number of predicted open reading frames (ORFs) changes. Also, a fully sequenced genome can be misleading because sometimes repetitive DNA is not completely sequenced. Therefore conservative terms are used to describe the results of genome analysis. Advances in knowledge that change our understanding of genomes include experimental determination of the protein function or structure of hypothetical proteins from uncharacterized ORFs, discovery of new types of noncoding RNA, and identification of new or unusual gene families.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.2

6) Explain the terms ″core genome″ and ″pan genome″ and describe how each contributes to the genome of a bacterial species. Give an example of genes that are part of a core genome versus those that are more often in the pan genome.
Answer: A core genome is the genome content present in all strains of a bacterial species. These are typically fundamental metabolic genes that combine to help define the bacterial species as a whole. The pan genome is the genome content that is found in only some of the strains within the species. The additional genes in the pan genome are what distinguish strains from one another. Typical genes in the pan genome may be for accessory functions such as virulence, degradation of a variety of substrates, or symbiosis. Often the pan genome contains mobile elements and plasmids.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.7

7) Interpret the genomic content of mitochondria in relation to their evolution. How is mitochondrial evolution more complicated than expected?
Answer: Some mitochondria contain accessory plasmids in addition to chromosomal DNA. The shapes of mitochondrial genomes are not all the same; some are linear and others are circular. These fundamental differences in mitochondrial genomes further complicate even basic comparative analyses. Some mitochondrial genes have also been transferred to the nuclear genome.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.4

8) Design an experiment using -omic methods to test how *Escherichia coli* adapts to different growth temperatures.
Answer: One could grow *E. coli* in identical flasks and media in two different temperatures (25 and 37 °C, for example). Then, you compare gene expression and protein content in the two cultures using transcriptomics and proteomics. Transcriptomics would require the extraction of RNA and the creation of cDNA, then sequencing and assembly of the transcripts to the genome for differential gene expression comparisons. For proteomics, the proteins would be separated from other cell components and then analyzed via 2D polyacrylamide gel electrophoresis. Quantified proteins matched to the proteome would be used to identify differentially translated proteins. (An alternative answer could describe use of a gene chip for gene expression analysis instead of sequencing of cDNA or quantifying peptides with HPLC-ESI-MS/MS instead of 2D-PAGE. Diagrams of the experiment should be encouraged.)
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.9, 9.10

9) Recommend what type of data should be collected and analyzed in a systems biology approach to investigate how a non-pathogenic bacterial strain becomes pathogenic. Describe what scientific fields and methods would be involved in your recommendation.
Answer: A systems biology approach integrates multiple methods and fields to model or analyze a problem. One approach would be to compare the genome and the proteome of non-pathogenic and pathogenic strains of a particular species to find the genes and proteins that are different and may be related to pathogenicity. This research would involve 3rd or 4th generation sequencing, 2D polyacrylamide gel electrophoresis of proteins, and bioinformatics to process and analyze all the data. Further experimentation could involve genetic manipulations to investigate the effect of removing or mutating the implicated genes and biochemical characterization of the implicated proteins. (Answers could vary significantly, but any reasonable discussion of how various ″-omic″ techniques and methods could be applied to this problem could be acceptable as long as the discussion is consistent and logical, linking the appropriate data types and methods to detect differences between pathogenic and nonpathogenic organisms. The answer should include the definition of systems biology.)
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.10

10) Chromosomal islands are presumed to have a ″foreign″ origin based upon three observations. What are these observations and how do they indicate that chromosomal islands are ″foreign″ in origin?
Answer: The presence of a sequence flanked by inverted repeats suggests transposition brought in the sequence. An island is also likely ″foreign″ when there is a dissimilar composition of bases (GC%) or a different codon bias. Lastly, chromosomal islands typically contain genes that are related in function, such as virulence or biodegradation.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.7

11) Explain why organisms undergoing rapid evolutionary change often contain relatively large numbers of mobile DNA elements, whereas once organisms settle into a stable evolutionary niche, most of these mobile elements are lost.
Answer: Genome diversity is likely to be high during a process of rapid evolutionary change. To obtain this high diversity, several processes, such as horizontal gene transfer and chromosomal rearrangements (e.g., deletions, inversions, translocations), are common. Organisms with a more stable niche require less diversity in their genomes and therefore tend to be less dynamic in genome content.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.6

12) The intracellular parasite *Nanoarchaeum equitans* has one of the smallest prokaryotic genomes. Why is the genome of *N. equitans* so small? How is it possible that the genome of *N. equitans* contains more genes than that of *Mycoplasma genitalium*, which is actually 90 kbp larger?
Answer: The intracellular parasite, *N. equitans* is a highly specialized archaeon with a very condensed genome, containing little noncoding DNA and no introns. It is well-adapted to live within its archaeon host and can hijack its host′s proteins. It therefore does not require its own genes for energy consumption. *N. equitans* contains more genes that *M. genitalium* because there is very little noncoding DNA and can thus fit more ORFs into a smaller genome. This is very efficient, although it does not allow for changes in gene expression in response to environmental changes.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.3

13) Explain why larger genomes contain more genes devoted to regulation than smaller genomes and why these genes increase the competitiveness of organisms.
Answer: Answers will vary, but one explanation is an increase in regulatory networks of larger genome-containing organisms provides a much greater opportunity for the cell to be dynamic in response to environmental stimuli. For example, this can be advantageous for cells to invoke certain pathways when particular carbon substrates become available. In this way, a cell can more finely tune its metabolic activities to minimize energy consumption and maximize energy acquisition.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3

14) Explain why the proteome is NOT defined as ″all the proteins encoded by an organism′s genome.″
Answer: Certain proteins might not be synthesized under a particular condition. Gene expression changes under different environmental conditions. Recognizing that not all potential ORFs will encode for proteins, it becomes difficult to determine *all* of the proteins within an organism, thus the proteome is only the proteins expressed under specific conditions.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.8

15) When analyzing the sequence of genes similar to ones already known, why is the amino acid sequence of the protein more important than the DNA sequence?
Answer: Over evolutionary time, a nucleotide sequence can change, yet its encoded protein function can remain the same. Individual nucleotide changes especially at the third position of a codon might not change the amino acid sequence at all. Depending on the location of the amino acid, some can substitute for others and not affect the protein′s function. Therefore a nucleotide sequence could be quite different from another, yet their function could remain the same.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.10

16) What process allows evolution to ″experiment″ with genes to create novel functions? How is this manifested in extant genes (presently existing) and their relationships to each other?
Answer: Duplication of genes through duplication of the entire genome or just parts of the genome allows the second gene to evolve and diversify into a different functional protein while still maintaining the original protein′s function. Gene duplication creates paralogs, which can further diverge through random mutations into several orthologs. Genes with shared evolutionary ancestry are called homologs (both paralogs and orthologs are homologs). Contemporary genes thus display relationships such as gene families, which are groups of homologous genes that may have different functions but share evolutionary ancestry.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 9.5

17) Explain how mass spectrometry (MS) has aided in the progression of the field of metabolomics.
Answer: Due to the complex nature of all possible chemical structures, metabolomics requires a highly sensitive instrument to identify each individual metabolite with high accuracy and precision. The field of metabolomics has greatly increased due to powerful MS approaches being used today. The capabilities of MS instruments such as a MALDI-TOF-MS permit the identification of many metabolites from a sample with high mass resolution when compared to older methods using NMR.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.11

18) The Human Microbiome Project is a large research program that aims to understand all of the microorganisms on and in the human body. Which ″-omic″ methods could be applied directly to a sample from the human body to study the microorganisms in the sample? Propose a general experimental approach for analyzing a sample containing a complex mixture of microorganisms.
Answer: A sample from the human body, such as stool or tooth swab, would be collected and the DNA or RNA extracted. RNA would be converted to cDNA. The cDNA would then be sequenced. This approach is metagenomics (and metatransciptomics for RNA). After sequencing, the short sequences must be assembled into longer stretches (genes or whole genomes). An alternative answer could outline the procedure of single-cell genomics from cells obtained from samples from the human body or the use of a gene chip for identification of organisms. Any answer that shows understanding of general steps and uses terms appropriately and consistently throughout the answer would be acceptable.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 9.4

19) Summarize the evolution of sequencing technology beginning with the Sanger method.
Answer: The first generation of DNA sequencing used the Sanger method to determine the DNA sequence. The Sanger method required that labeled dideoxynucleotides were added to a reaction so that the elongation of new DNA chains would terminate when the dideoxynucleotides were incorporated. In first generation sequencing, the detection of the terminated chains was not very sensitive. Second generation sequencing used similar chemistry, but miniaturized the reactions and improved detection methods and computation so that more sequence information could be collected very rapidly. Third generation sequencing improved detection such that single molecules could be detected, thus even more sequence could be generated. Fourth generation sequencing has moved beyond optical or light detection by using pH or charge changes to detect single molecules.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.2

20) Explain why parasitic and endosymbiotic organisms such as Tremblaya and Hodgkinia can have unusually small genomes and what genes they may lack.
Answer: These organisms coexist with another organism and are able to use the enzymes that their host produces to perform metabolic functions. In some cases, they rely on their hosts for both catabolic and anabolic enzymes.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 9.3